

Welcome to GNS 103: Information Retrieval Blog

COURSE OUTLINE

☒ Lesson 1: Introduction to Libraries

1. Definition of library
2. Types of library – Private/Public/National/School/Academic (university, college and polytechnic libraries)/ Digital, electronic, virtual libraries/Archives
3. Functions/Services of University Library to users
4. Introduction to FUTA Library

☒ Lesson 2: Forms by which is recorded

1. Primary, Secondary and tertiary sources
2. Print forms
3. Non-print forms/ audiovisual resources
4. The Book: definition, Physical and Bibliographic parts

Purpose of the course

- To create awareness about information resources in the library
- To create awareness about the different methods of accessing information resources in the library
- To expose students to information resources outside the learning environment
- To develop users' study and research skills
- To develop the information literacy skills of the students.

Information literacy is an important objective of the course. The American Library Association (**ALA**) states, "To be information literate, a person must be able to recognize when information is needed and has the ability to locate, evaluate, and use effectively the needed information."

Lesson 1: Introduction to Libraries

Definition of library

A library is an institution responsible for the acquisition, organisation and storage of recorded knowledge in various media for study, research and consultation. Libraries have come a long way. From the early days when writing was done on cuneiform, papyrus and parchment to when scribes and monks used to manage libraries consisting of handwritten manuscripts to when **Johannes Gutenberg introduced printing in 1445**, leading to the evolution of modern libraries. Libraries have served humanity over the years. Today a library contains print, non-print and electronic resources.

Types of library

- i. **Academic libraries:** Attached to post secondary or tertiary institutions. The sub types are University libraries, College libraries and Polytechnic libraries. The perform functions directly related to the mission and goals of their parent institutions.

The university library is unique in a number of ways. A university is established for research, teaching and community development, and the university library supports these goals. The users of a university library include undergraduate and graduate students, lecturers or academic staff and researchers. Academic libraries operate long opening hours. Some of the important activities of academic libraries are inter library corporation, teaching of library skills, lending and borrowing and collection of locally published (university-wide) materials /depository.

- ii. **Public libraries:** The use of libraries is unrestricted; it serves the needs of all categories of users. State libraries and community information centres are examples of this type of libraries. They are usually established by government act. The government established boards or NGOs may manage them. They support the information, educational and recreation needs of the members of the community. Public libraries serve as a cultural centre as they are usually venues for public events. They champion adult literacy programmes and continuing education activities. They are however grossly underfunded.

- iii. **National libraries:** Each country owns a national library that handles national bibliographic issues. It is also called the apex library. For example, there is the National Library of Nigeria, The British Library and the Library of Congress of the United States. The roles of the National library include:

- ☒ Protecting copyright interests of authors or publishers
- ☒ Issuing ISBN and ISSN
- ☒ Accepting legal deposit -which are free copies of published materials required by law to be sent to the national library by publishers or authors
- ☒ Publication of the national bibliography: a list of all the books published in the country within a particular year
- ☒ Providing CIP service: Cataloguing in publication service
- ☒ Advising the government on bibliographic issues among other services

- iv. **School libraries:** These libraries are attached to pre-primary, primary and secondary schools to serve the needs of children and young adults. They support teaching of school children and are heavily dependent on audiovisual resources like pictures, audio recordings etc. They are also called media resource centres.

- v. **Special libraries or research libraries and documentation centres:** They are owned or affiliated to professional associations, government departments, companies and research institutes. They provide specialized data and technical information to enable the personnel of the institutions carry out their duties. National Root Crop Research Institute Library, IITA Library are examples.
- vi. **Archives and record management centres:** These institutions that collect and preserve public and historical records for posterity. E.g. National Archives at Ibadan
- vii. **Electronic libraries:** (i) Virtual libraries: These libraries exist in space (on the internet) and not physically. (2) Digital libraries are computer-based but may not be on the internet or online. Information may exist on CD-ROM databases and other digital databases.
- viii. **Private Libraries:** Private individuals who develop the collection in line with their interests own these. Dr Azikiwe and Chief Awolowo owned personal libraries, which they bequeathed to institutions.

Functions/Services of University Library to users

- ☒ Lending services
- ☒ Inter library loans
- ☒ Reserved services
- ☒ Sitting and studying space
- ☒ Reference services
- ☒ Referral services
- ☒ Current Awareness Services
- ☒ Selective Dissemination of Information
- ☒ Exhibition and displays
- ☒ Library publication
- ☒ User education
- ☒ Literature search
- ☒ Video recording and photography
- ☒ Translation services
- ☒ Rental of facilities and space
- ☒ Extension services

Introduction to FUTA Library

- ☒ Established in 1982 and opened in 1983 to assist in research, learning, teaching and community service missions of the university.
- ☒ Moved to the a three-floor complex in the main campus in 2006

- ☒ 3 floors: reading rooms, administrative offices, workrooms
- ☒ Units / departments: Circulation, Cataloguing and Classification, E-resources, Reference, Serials, bindery, security, Multimedia (Audio visual), Acquisition, Special Collections (BRD).
- ☒ Membership to the library is free and open to all academic and senior staff and students of the university.
- ☒ Only bona fide students of the university who have registered in the university library are allowed into the library. They must possess a student's ID card for identification and renew their membership at the beginning of every session.
- ☒ Hours of service
Monday through Friday -8.00am to 9.00pm
Saturday – 9.00am to 6.00pm
Sunday – 4.00pm to 9.00pm
Public Holidays- Closed
During Vacation: Monday through Friday 8.00-4.00 pm , closed during the weekends

Lesson 2: Forms in which Knowledge is recorded

Whether they are:

- ☒ **Primary resources:** original resources not yet interpreted or condensed e.g. Research reports, journal articles, newspaper articles, thesis
- ☒ **Secondary resources:** Condensed , interpreted or analyzed information like dictionaries, textbooks, handbooks or
- ☒ **Tertiary resources:** used to locate other sources of information e.g. bibliographies, indexes etc, : Library resources can exist as print, non-print or electronic forms

Print forms: Print forms include

- ☒ References sources – resources that are for consultation and not continuous reading e.g. dictionaries
- ☒ Government publications – issued by government agencies e g. Hansard
- ☒ Periodicals/serials – issued periodically at regular intervals e.g. journals
- ☒ Monographs – a publication that addresses a single topic
- ☒ Pamphlets – printed material below 50 pages

The Book:

A book is a document of above 49 pages, bound together at one edge and possessing a distinctive title.

- ☒ Physical parts of a book:
Binding or cover,
Spine

Contents

- ☒ Bibliographic parts of a book consists of the preliminary pages and the main text
 - (a) Preliminary pages
 - Fly leaves, which are blank
 - Half title page contains the (abbreviated) title
 - Title page contains full title and author/s name/s, may contain other bibliographic Information
 - Copyright page: (Verso page) Contains the copyright information (copyright owner, ISBN, publishing history, date of publication, place of publication, publisher)
 - Dedication page
 - Preface: provides the reason for writing the book
 - Table of Contents: an outline of the contents of the book
 - Acknowledgment – to acknowledge those who have assisted with the work
 - (b) Main text
 - Text
 - References and bibliography: a list of the sources consulted in the preparation of the book.
 - Appendix: contains supplementary information added to the book
 - Glossary: a list and definitions of difficult terms used in a book, found at the end of the text
 - Index: Alphabetical list of the terms used in a book with their page number to indicate where they are located

Non-print forms / non-book materials or Audio-Visual resources: Do not exist as print, and include broadly:

- ☒ Cartographic materials – maps
- ☒ Graphic materials_ pictures, drawings, real objects, charts, transparencies
- ☒ Manuscripts – handwritten documents
- ☒ Audio recordings – audiotapes, records
- ☒ Motion pictures and video recordings – filmstrips
- ☒ Microforms – microfilms, microfiches, micro cards. Micro-record of print information miniaturized by photographic processes
- Electronic materials which may be found as: Digital materials like- CDs, DVDs or online/virtual

Non-book materials are valuable sources of information and contain huge amounts of data while occupying little space. They serve well for conservation of information. For example, valuable records can be microfilmed; a strip of microfilm can contain several pages of information. They aid retention because non –print resources appeal to

learners' senses of sight and sound, they create vivid impressions on their minds that cannot easily be forgotten.

However, most non-print medium require machines or hardware for their use e.g. a projector or micro-reader and because the hardware are bulky, they cannot be carried about with ease. They also require special preservation methods if they are to last (Special temperature and humidity levels). Power supply is also required to make most of the hardware work.

Lesson 3: Reference Tools and Services

1. Basic/background information sources- encyclopedia, dictionaries
2. Fact/ready reference sources – directories, handbooks, yearbooks
3. Literature search tool- abstracts, indexes, bibliographies
4. Geographical sources – atlases, gazetteers, travel guides
5. Biographical sources – “who’s who”, biographies
6. Electronic sources: online reference sources, databases

Lesson 3: Reference Sources and Services

Reference sources are library resources intended for consultation only and not continuous reading. They are not shelved with the regular collection and supply particular pieces of information. Reference staff use them to answer directional, ready reference, search and research queries. They may also exist as digital (CD-ROM) or online forms.

Other characteristics of reference resources:

- ☒ Alphabetical or chronological arrangement of information for easy retrieval
- ☒ They do not circulate (cannot be borrowed)
- ☒ May exist as multiple volumes
- ☒ May be general or specialized
- ☒ They are marked with “REF” to indicate their location. This is inscribed right under the call number and helps to differentiate it from other materials.

For convenience, we can classify these resources into the following groups.

Basic/background information sources: contain basic information on subjects

1. **Encyclopedia:** An encyclopedia (also spelled encyclopaedia or encyclopædia) gives a broad overview on either all branches of knowledge (a general encyclopedia) or a particular branch of knowledge i.e. (subject encyclopedia). They are regarded as the backbone of reference service. Encyclopedia entries are arranged alphabetically by

article name. Encyclopedia entries are longer and more detailed than those in most dictionaries, unless for encyclopaedic dictionaries. Unlike dictionary entries, which focus on linguistic information about words, encyclopedia articles focus on information to cover the subject/s. It is a good starting point for research. An example of a general encyclopedia is the Encyclopedia Britannica which consists of 32 volumes including 13 volumes of Micropeadia or ready reference, 16 volumes for Macropedia , 1 volume of Propedia (Outline of Knowledge) and 2 volumes of index. An e.g. of a subject encyclopedia is the McGraw Hill Encyclopedia of Science & Technology. Encarta and Wikipedia are examples of online encyclopedia. Encyclopedia are revised or updated as supplements or yearbooks.

2. Dictionaries: A dictionary is an alphabetical listing of words with their definitions and other features such as usages, etymologies (history of the word), derivation, syllabification, phonetics, pronunciations, and other information. Dictionaries are classified as general-purpose dictionary, subject dictionary and special purpose dictionary.

- ☒ General purpose dictionary: Oxford English dictionary
- ☒ Subject dictionary – Dictionary of Engineering, Dictionary of genetic Engineering
- ☒ Special purpose dictionary – Dictionaries of slang, etymology (history of words), thesaurus, bilingual or dual purpose dictionaries, Oxford dictionary of quotations, Webster’s Thesaurus, Dictionary of slang

Dictionaries may be abridged (pocket dictionaries), encyclopaedic (having a wide scope) or desk dictionary depending on its scope or size. Other variations of dictionaries are:

- ☒ **Lexicon** : an alphabetical list of the words giving the vocabulary of a particular subject area
- ☒ **Glossary**- usually attached to the end of a book, a list of difficult terms in the book and their definitions in that context.

Fact/ready reference sources

These are required for the commonest queries raised by the user e.g. “Who is the president of Norway?” or “What is the capital of Finland”?

3. **Directories:** This is an alphabetical list of residents, organizations, businesses or professionals in a geographical area and their contact information. They are used to locate or verify (the names) and addresses of individuals , organizations and institutions. It also includes their addresses, affiliations, telephone numbers and other contact information. Some examples of directories are:

- ☒ Directory of directories
- ☒ Yellow pages: trade directories like a directory of publishers or automobile engineers

- ☒ Business directory
- ☒ Educational directory: World of learning, World List of Universities
- ☒ Telephone directories

4. **Handbooks**, Manuals/ How to books and guides: Provides information on all aspects of a given subject in a concise manner. It is usually written for practitioners, they serve as a handy guide to a particular subject. Large amounts of information or knowledge on subjects are compressed into a smaller volume/ volumes. They are usually heavily illustrated. Manuals are especially useful in explaining how to operate machines E.g. The Smartcard Handbook, Macmillan Guide to Trees etc.

5. **Yearbooks and almanacs**: A yearbook is an annual record that highlights, and commemorates the past year of a school or institution. The term may also refer to a book of statistics or facts published annually. **An almanac** is an annual publication and includes a comprehensive presentation of statistical and descriptive data covering the entire world or some geographical area, discussions of topical developments and a summary of recent historical events. E.g. Whitaker's Almanack, TIME Almanac with Information Please, World Almanac and Book of Facts, Guinness Book of Records, Statesman's Yearbook. Other major topics covered by almanacs are geography, government, demographics, agriculture, economics and business, health and medicine, religion, mass media, transportation, science and technology, sport, and awards/prizes.

Literature search tools are bibliographic retrieval tool used to identify relevant information sources. They can be used to build up a collection and verify incomplete or inaccurate data.

6. **Indexes**: An index is a guide to all documents relevant to a particular subject irrespective of the format. The bibliographical information is arranged alphabetically, giving full bibliographic information. Types of indexes include:

- ☒ Book indexes- found at the end of books
- ☒ Periodical indexes – for periodicals like magazines
- ☒ Newspapers indexes – created for newspapers
- ☒ Citation indexes- that appear at the end of journal articles

7. **Abstracts**: An abstract can be referred as a true surrogate of a document. An abstract is different from an index in that it offers a summary alongside bibliographical information. It is often used to help the reader quickly ascertain the paper's purpose and relevance. Abstracts may be merely descriptive or evaluative. E.g. Dissertation Abstract International, Chemical Abstracts

8. **Bibliographies**: A bibliography means (Information about books- derived from the Greek words "biblio" and "graphien". It may mean cited sources at the end of a book but

as separate works, they may be in bound volumes. They list in systematic order (alphabetically, chronologically or by format) materials relevant to a particular subject. They supply bibliographical data –author, title ,date of publication, etc. of these materials Bibliographic works differ in the amount of detail provided depending on their purpose, and can be generally divided into two categories: enumerative bibliography, which results in an overview of publications in a particular category, and analytical or critical bibliography. Other types of bibliography are: national bibliography, trade bibliography or publisher's catalogue, bibliography of bibliographies and subject bibliographies. Bibliographies may also be current or retrospective depending on the time span it covers

Geographical sources: give information about places and physical locations like towns, cities, geographical features e.g. 'Where are Haiti'? 'What is the height of the Kilimanjaro"

9. **Atlases:** An atlas is a collection of maps; it is typically a map of Earth or a region of Earth, but there are atlases of the other planets (and their satellites) in the solar system. Atlases have traditionally been bound into book form, but today many atlases are in multimedia formats. In addition to presenting geographic features and political boundaries, many atlases often feature geopolitical, social, religious and economic statistics and current affairs. .Eg. Times Atlas of the World, Atlas of Nigeria

10. **Gazetteers:** A gazetteer is a geographical dictionary used in conjunction with a map or a full atlas. It typically contains information concerning the geographical makeup of a country, region, or continent as well as the social statistics and physical features, such as mountains, waterways, or roads. Examples of information provided by gazetteers include the location of places, dimensions of physical features, population, GDP, literacy rate, etc. Arrangement is alphabetical. Example: Columbia Gazetteer of the World, Merriam-Webster's Geographical Dictionary.

12. **Travel guides or Traveller's guidebook** is a book for tourists or travellers that provide details about a geographic location, tourist destination, or itinerary. It is the written equivalent of a tour guide. Many travel guides now take the form of travel websites rather than printed books. It will usually include details such as phone numbers, addresses, prices and reviews of hotels and other lodgings, restaurants, places of interest and activities. Maps of varying detail are often included. Sometimes historical and cultural information is also included. E.g. Nigerian Hotel and Travel guide. Biographical sources give information about people living or dead

13. **Who's Who or Who is Who** are biographical dictionaries or directories. They contain concise biographical information on a particular group of people e.g. medical doctors. They respond to queries such as "Who is Lord Lugard?" "How many books has Chinua Achebe written?". It may be current or retrospective. A retrospective version is about

dead or deceased people and is called "Who was Who" .

14. **Biographies:** A biography is a detailed description or account of someone's life and the times. A biography includes details like education, work, relationships, a biography presents a subject's story, highlighting various aspects of his or her life, including intimate details of experiences, and may include an analysis of the subject's personality. An autobiography is written by the subject himself.

Lesson 4: Serials

1. Periodicals
2. Newspapers and magazines
3. Annuals
4. Journals

Lesson 4: Serials /Periodicals

Serial materials, as sources of information are unique because of the following characteristics:

- ☒ Issued periodically at regular intervals it could be daily, weekly, monthly annually etc.
- ☒ They are up-to-date and current
- ☒ Supplements to book materials
- ☒ Issued as volumes and numbers
- ☒ Content are referred to as articles, not chapter
- ☒ ISSN (International Standard Serial number) is the standardized reference number used for serial materials of all types
- ☒ Index to periodicals are used to search for information in periodicals
- ☒ Catalogue cabinet used for serial material is called kardex.

Some notable search tools for periodicals Ulrich International Periodical Directory, Nigerian periodicals index, newspaper indexes e.g. Times index and citation indexes for journals. Newspapers and magazines contain information for a more general public, while journals are for a more specialized audience (professionals) are good sources for research.

THE INTERNET

INTERNET TUTORIALS: your basic guide to the internet
Understanding the World Wide Web

This tutorial covers the basics of the World Wide Web, focusing on its technical aspects. After all, the Web is a technological phenomenon. Therefore it's useful to

understand some of the fundamentals of how it works.

The world wide web is a system of Internet servers that supports hypertext and multimedia to access several Internet protocols on a single interface. The World Wide Web is often abbreviated as the web or www.

The World Wide Web was developed in 1989 by Tim Berners-Lee of the European Particle Physics Lab (CERN) in Switzerland. The initial purpose of the Web was to use networked hypertext to facilitate communication among its members, who were located in several countries.

Protocols of the Web

The surface simplicity of the Web comes from the fact that many individual protocols can be contained within a single Web site. Internet protocols are sets of rules that allow for intermachine communication on the Internet. These are a few of the protocols you can experience on the Web:

HTTP (HyperText Transfer Protocol): transmits hypertext over networks.

E-mail (Simple Mail Transport Protocol or SMTP): distributes e-mail messages and attached files to one or more electronic mailboxes.

FTP (File Transfer Protocol): transfers files between an FTP server and a computer, for example, to download software.

VoIP (Voice over Internet Protocol): allows delivery of voice communications over IP networks, for example, phone calls.

Hypertext and links: the motion of the Web

The operation of the Web relies primarily on hypertext as its means of information retrieval. HyperText is a document containing words that connect to other documents. These words are called links and are selectable by the user. A single hypertext document can contain links to many documents. In the context of the Web, words or graphics may serve as links to other documents, images, video, and sound. Links may or may not follow a logical path, as each connection is created by the author of the source document. Overall, the Web contains a complex virtual web of connections among a vast number of documents, images, videos, and sounds.

Producing hypertext for the Web is accomplished by creating documents with a language called hypertext markup language, or html. With HTML, tags are placed within the text to accomplish document formatting, visual features such as font size, italics and bold, and the creation of hypertext links.

Pages on the Web

The backbone of the World Wide Web are its files, called pages or Web pages, containing information and links to resources – both text and multimedia – throughout the Internet.

Access to Web pages can be accomplished in all sorts of ways, including:

1. Entering a Web address into your browser and retrieving a page directly
2. Browsing through sites and selecting links to move from one page to another both within and beyond the site

3. Doing a search on a search engine to retrieve pages on the topic of your choice (See: The World of Search Engines)
4. Searching through directories containing links to organized collections of Web pages (See: The World of Subject Directories)
5. Clicking on links within e-mail messages
6. Using apps on social networking sites or your mobile phone to access Web and other online content
7. Retrieving updates via RSS feeds and clicking on links within these feeds (See: RSS Basics).

Retrieving files on the Web: the URL and Domain Name System

URL stands for uniform resource locator (or global address sic). The URL specifies the Internet address of a file stored on a host computer, or server, connected to the Internet. Web browsers use the URL to retrieve the file from the server.

Underlying the functionality of a URL is a base numeric address that points to the computer that hosts the file. This numeric address is called the IP (internet protocol) address. The host portion of a URL is translated into its corresponding IP address using the domain name system (DNS).

For example, the DNS translates <http://www.microsoft.com> into the IP address 207.46.19.254.

Every file on the Internet, no matter what its protocol, has a unique URL. Each URL points to a specific file located in a specific directory on the host machine. This is the format of a URL:

protocol://host/path/filename = <http://www.senate.gov/general/capcam.htm>

This URL is typical of addresses hosted in domains in the United States. The structure of this URL is shown below.

1. Protocol: http
2. Host computer name: www
3. Second-level domain name: senate
4. Top-level domain name: gov
5. Directory name: general
6. File name: capcam.htm

Domains

.com commercial enterprise

.edu educational institution

.gov U.S. government entity

.mil U.S. military entity

.net network access provider

.org usually non-profit organization

.ng country code for Nigeria

.uk country code for United Kingdom

Applications (apps)

Applications, commonly called apps, are small programs that run within various online environments. These programs allow you to enjoy functionalities that enhance your experience within that environment.

Social networking sites often make use of apps. For example, Facebook is well-known for featuring thousands of apps created by Facebook or outside developers. These apps allow you to play games, shop, form issues-based communities, find family or classmates, etc.

Source: <http://www.internettutorials.net/www.asp>

Other Internet terms

Websites: is a set of related web pages containing content such as text, images, video, audio, etc. A website is hosted on at least one web server, accessible via a network such as the Internet or a private local area network through an Internet address known as a Uniform Resource Locator. All publicly accessible websites collectively constitute the World Wide Web.

Web Browsers: A web browser is an application software or programme designed to enable users access, retrieve and view documents and other resources on the internet. E.g. Mozilla Firefox, Google chrome, Internet Explorer, Opera and Safari.

Web Search engines : is designed to search for information on the World Wide Web. The search results are generally presented in a line of results often referred to as search engine results pages e.g Google, Yahoo, Alta Vista, Excite, Ask.com, Maama, Devil finder

Discussion groups: an online forum for individuals to discuss various topics amongst each other. People add their comments by posting a block of text to the group. Others can then comment and respond. Discussion groups include web-based forums, bulletin boards, listservs, electronic mailing lists, and newsgroups.

Social Media Networks: refers to the means of interactions among people in which they create, share, exchange and comment contents among themselves in virtual communities and networks. E.g. Wikipedia, blogs, Facebooks, twitter, Linked In, Youtube

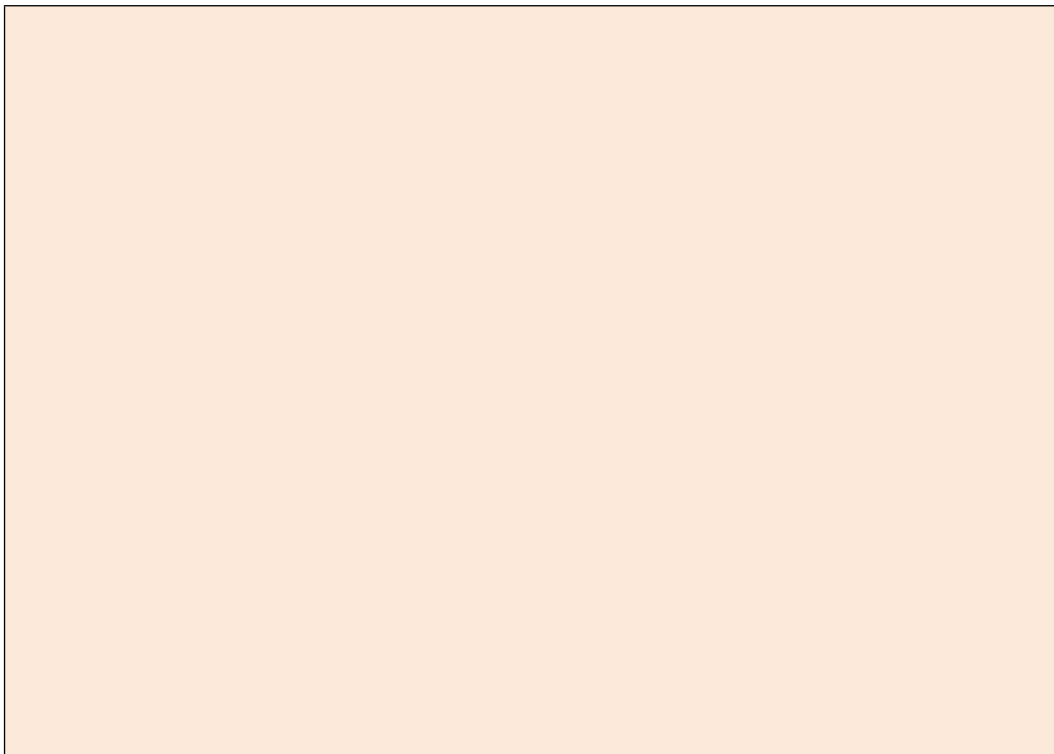
Blogs: is a discussion or informational site published on the World Wide Web and consisting of posts typically displayed in reverse chronological order the most recent post appears first. Class blogs and wikis: There are a variety of Web 2.0 tools that are currently being implemented in the classroom. Blogs allow for students to maintain a running dialogue, such as a journal, thoughts, ideas, and assignments that also provide for student comment and reflection. Wikis are more group focused to allow multiple members of the group to edit a single document and create a truly collaborative and carefully edited finished product.

ORGANIZATION OF KNOWLEDGE

Lesson 5: Organization of Knowledge

- Classification schemes: The purpose of classification /Different schemes

- i. Library of Congress Classification Scheme
- ii. Dewey Decimal Classification scheme
- iii. Universal Decimal classification scheme
- Forms of Catalogue
 - i. Book catalogue
 - ii. Card catalogue
 - iii. OPAC
 - iv. Machine Readable Catalogues (MARC)
- Types of catalogue
 - i. Dictionary
 - ii. Divided
 - iii. Classified catalogue
 - iv. Union Catalogue
- Typical catalogue card



Organization and retrieval of knowledge in the library – Classification Schemes

The principle of organisation is to arrange library resources in a manner that will

facilitate easy access and retrieval when users need these resources. Classification implies that knowledge is organised in a systematic order of subjects. Classification ensures that library materials with similar subjects are kept together.

Library Classification Schemes

1. Library of Congress Classification Scheme: Knowledge is broken into 21 classes under this scheme from A-Z. It is an alphanumeric scheme – a mixed notation consisting of alphabet(s) and numbers. Each class is also subdivided to capture all aspects of the subject in question.

Relevant Classes

A-General works

G – Geography

H – Social science (and Management)

N- Fine Arts

Q- Science

QA- mathematics

QB – Astronomy

QC – Physics

QD – Chemistry

QE – Geology

QH – Natural History and Biology

QK – Botany

QL – Zoology

QM – Human anatomy

QP – Physiology

QR- Microbiology

S- Agriculture

T- Technology

TA – Engineering general/ Civil engineering

TC- Hydraulics engineering, harbours, rivers, canal

TG – Bridge and roof engineering

TH – Building, fire prevention and extinction

TN- mining engineering

TK – Electrical engineering and industries

R- Medicine

Z- Library Science/ Bibliography

Division and sub-division:

T- Engineering

TN- Mining engineering

263 mineral deposits or 270 prospecting

Class mark for a book on mineral deposits will be TN 263

Dewey Decimal Classification Scheme (DDC) use number notations and the decimal number system. DDC is used in small libraries and divides knowledge into 10 broad classes ranging from 000-999. DDC was introduced by American librarian and library educator Melvil Dewey in 1873 while he was a student at Amherst College in Massachusetts. DDC is expansive (can accommodate many subject classes and sub-classes) and also hierarchical in the relationship of these classes and subclasses. The ten primary classes are as follows:

000 Generalities

100 Philosophy and psychology

200 Religion

300 Social sciences

400 Language

500 Natural sciences and mathematics

600 Technology (applied sciences)

700 The arts; fine and decorative arts

800 Literature and rhetoric

900 Geography and history

The Dewey Decimal Classification is based on multiples of ten. Thus, each of the ten main classes has the potential to be broken down into smaller multiples of ten. For example, class 500 (natural sciences and mathematics) has the following subclasses:

500 Natural sciences

510 Mathematics

520 Astronomy and allied sciences

530 Physics

540 Chemistry and applied sciences

...

Each of these classes may then be further divided. For example, the subclasses of 540 are as follows:

540 Chemistry and applied sciences

541 Physical and theoretical chemistry

542 Techniques, equipment, and materials

543 Analytical chemistry

544 Qualitative analysis

...

548 Crystallography

549 Mineralogy

Each of these classes may be divided ten additional times and those ten more times, and so forth. At each step of the hierarchy, one additional number from 1 to 9 is added to the length of the notation. Thus, class 500 (natural sciences and mathematics) has subclass 540 (chemistry and applied sciences); a subclass of 540 is 541 (physical and theoretical chemistry), a subclass of 541 is 541.3 (physical chemistry), and so on.

Universal Decimal Classification

Another major classification system is the Universal Decimal Classification (UDC) and based on the Dewey Decimal Classification, the Universal Decimal Classification combines notation to express multiple concepts. For example, 940(=395) indicates a work on the history of Vikings in Europe—940 is the top-level notation designating a work on the history of Europe, and 395 is the bottom-level notation denoting a study of Vikings. The major purpose of the Universal Decimal Classification is to identify the content of documents.

The Catalogue

Library resources or information carriers such as books, files, periodicals and even audiovisual resources need to be made available or accessible to users and this is done through retrieval tools like the catalogue.

The catalogue is a list of the library holdings. The catalogue serves the following functions:

- ☒ It provides a complete bibliographic list of every item in the library
- ☒ To identify and locate a particular document or item in a collection
- ☒ To bring together all related material in a collection
- ☒ To enable the user to evaluate and select relevant titles among many entries
- ☒ It describes the collection
- ☒ It provides author, edition and publication (bibliographic detail) information
- ☒ It provides subject information

Types of catalogue

This refers to the ways in which a catalogue may be arranged:

- i. **Dictionary Catalogue:** the catalogue records here are arranged in one long alphabetical (A-Z) order that includes author, title and subject.
- ii. **Divided Catalogue:** This type catalogue divides the catalogue records into two. 1. Author and title and 2. Subject. They are also arranged alphabetically in each of the cabinets. It is flexible and easy to manage and use.
- iii. **Classified catalogue:** This catalogue is arranged according to the class mark.

- The Shelf list is a classified catalogue
- iv. **Union Catalogue:** This type of catalogue is shared between more than one libraries separated geographically. Each of the co-operating libraries has access to the catalogue and the contents of their various libraries.

The cataloguing process involves:

- ☒ Descriptive cataloguing: bibliographic data of the material like the author, publisher, place of publication, number of pages etc are indicated on the 5"by 3" card or other form of catalogue
- ☒ Subject cataloguing: Subjects that describe the content of the book are also indicated on the card. These subjects are derived from the subject heading
- ☒ Classification: Class marks from the schedules are used to identify the material. The class mark corresponds with the subject content.
- ☒ Assigning author mark or cutter number using the cutter author table.

The call number is a combination of the class mark and an alphanumeric (consisting of both letters and numbers notation, the author mark) that usually represents the surname of the author and a unique number assigned to that name or the initial letter of the first word of the title.

The cataloguing and classification process is carried out in FUTA library using these tools: The Library of Congress Subject Heading (LCSH), Library of Congress Schedule and the Cutter table.

Forms of Catalogue

- i. **Book catalogue:** Old, inflexible and outdated but cheap and portable. May still be used for a small collection
- ii. **Card catalogue:** Commonest catalogue form. The card is a 3"by 5"card, which is used to describe library materials in a collection, they (cards) are then arranged alphabetically to make retrieval easy. They are housed on trays and fixed in a cabinet systematically. Each book or library material may have several cards representing it because the catalogue card takes care of various access points: namely the author or authors, the title, the subjects, the class mark and the series if any. This is because the user who is not on the shelves yet may decide to check up a book from any of the access points. It is easy to maintain and flexible.
- iii. **OPAC:** Online Public Access Catalogue is a computerized catalogue system. A number of computer programmes e.g. SLAM, GLAS, X-LIB, ALICE for Windows, CDIS/ISIS are used to manage the database. In FUTA Library SLAM, which means

Strategic Library Automation Management is used to manage the OPAC. OPAC is modern, user friendly and efficient but may be expensive. It can be accessed remotely because it is a networked system, so it is highly accessible.

- iv. **Machine Readable Catalogues (MARC):** This catalogue may be stored on micro format or CD ROM.

Typical catalogue card and its elements

main entry catalogue card

A main entry card and added entry cards are made for each book to cover all the access points – author(s), series, title, subjects and class mark.

Use of Information and Communication Technology (ICT) in the Library (Library Automation)

Information and Communication Technology (ICT) has transformed library services globally. Most current information are recorded in electronic format, ICT has also contributed immensely to the performance of librarians in the discharge of their duties such as in cataloguing, reference services, circulation management, serials control etc. ICT has contributed to the library in the following specific ways.

Library management softwares

Libraries utilize softwares designed to manage different library routines and processes. Most of these softwares are integrated and have modules for the different activities or tasks carried out in the library like cataloguing, statistics, acquisition processes, serials control etc. Some examples of such softwares are CDS/ISIS, GLAS, ALICE for Windows, X-Lib and SLAM. SLAM is used in the University Library FUTA and stands for (Strategic Library Automation Management).

OPAC: This means Online Public Access Catalogue and is the computerized version of the library catalogue or a database of the library holdings. The advantage of the OPAC over manual methods is ease of use and the fact that it saves space. It provides access to the catalogues of a library on the local intranet, extranet or even the internet.

Office Operations: Word processing, accounting,, database management and communication through e-mail are all enabled in the library through ICT.

Networking: Library users can access information of various types such as online databases, e-journals, e- books, government publications digitally through networked systems. Access may be allowed online remotely through the internet or intranets.

Electronic Document Delivery: Libraries may not rely anymore on postal services to send documents to users or carry out interlibrary lending. Libraries send documents through electronic networks that can deliver documents in various formats e.g PDF straight to users' desktops.

Online user education or tutorials: Libraries can use the internet or CD –ROMS to educate their users or carry out information literacy programmes. Virtual tours can be offered online making user education more convenient for all.

E-reference services: Some services such as SDI (Selective dissemination of information) or Current Awareness Services (CAS) and virtual reference desks, announcements of new acquisitions and other reader advisory services can be made easier through the internet. Users can have online interaction with the reference staff.

Library cooperation and resource sharing: A central union catalogue can be better managed through ICT, thus libraries can create and share bibliographic records and other information resources in digital format.

Institutional repositories are publications that originate locally from within the university community such as theses, dissertations, reports, conference papers and seminar papers. ICT has made it possible not only to provide better access to these resources but also to ensure the preservation of the resources.

E- libraries: Digital libraries depend on information recorded on digital formats like CD-ROMS. Virtual libraries are libraries that do not exist in physical space or structure but can be accessed via networks. E.g. The Nigerian Virtual Library.

Social Media Networks: Social media networks like twitter, facebook and linkedIn, are some interactive internet services that are presently serving as communication forum for librarians and their uses. These networks can be deployed for educational uses. Discussion groups, listserves and communities also assist library services

E- mails: This is a means of communication between the library and the users.

Library websites: A medium of communication for libraries to their users. It is also used to promote the library and publicise it.

Online searching: searching of online databases like AGORA, ERIC. Browsing and surfing the internet through search engines, metasearch engines and subject directories to supplement library sources

Advantages of Using ICT in the library

- ICT makes library work easier, faster, cheaper and more effective.
- Helps to manage information overload as information retrieval is made easier in computerized systems.
- Remote access is enabled through networked systems
- Computerization saves space and reduces paper.

Challenges of using ICT in libraries

- Poor funding of ICT infrastructures
- Constant change of software and hardware
- Erratic power supply
- Insufficient bandwidth
- Lack of technical IT knowledge by library staff
- Copyright and intellectual property rights management

Search Strategy and Report Writing

A search strategy is a systematic plan for conducting a search. In fact, a little time spent in formulating a search strategy will both save the researcher/writer time and produce better results.

Steps in a search strategy

1. Define the problem by identifying the important concepts of the search (task definition). There may be need to consult subject dictionaries and encyclopedias and discuss the project with the supervisor to understand the problem better.
 2. Choose key words that describe the concepts , they may be synonyms, related terms or other variations of the keywords
 3. Using information retrieval tools (literature search tools) like abstracts , indexes, the catalogue or search engines, try to locate suitable materials on the assignment. The search should cover a variety of information sources – textbooks, websites, journals, newspapers, reference materials etc.
 4. Evaluate your search results for authoritativeness, currency and accuracy etc
 5. Prepare the references.
- The above steps also embody the information literacy skills required to solve information problems.

Report writing

Students may be required to produce written reports as part of their course. A report is an example of an information problem. A report aims to inform and has the following

characteristics:

- formal style
- introduction, body and conclusion
- analytical thinking
- careful proof-reading and neat presentation

A report has the following sections

- Title page
- Table of contents
- Executive summary/abstract
- Introduction
- Body
- Conclusion
- Recommendations
- References

Referencing

One of the most important aspects of academic writing is making use of the ideas of other people thus at the end of a report a writer must include a reference list. A reference list is an alphabetical list of all the sources actually quoted or cited in the document. Citations are used to refer the reader to another work mentioned or quoted inside the text of the report or dissertation to illustrate a point or confirm an idea. A bibliography on the other hand is an alphabetical list of sources or references on a particular topic. Bibliographies indicate further reading or additional information sources on the topic of the report and is included at the end of the text after the list of references.

There are a number of standard styles used to acknowledge research sources such as MLA, Chicago Manual of Style and American Psychological Association (APA) style which is the familiar method in the university.

Within the text, a book or article is cited by the author's surname and year of publication. It is very important when quoting any idea/writing to use the writers own words, unless you are quoting. It must be clear when the words or ideas being used are the writers and when they are taken from another writer.

Use of another person's words or ideas as if they were one's own is called plagiarism and is regarded as a very serious offence. In report writing the writer is expected to put forth his/ her own ideas, however sometimes it is necessary to cite others researchers or writers to show one's level of familiarity with the subject. it also guards against plagiarism. Referencing helps the reader follow up any work mentioned in the text if the reader is interested. The reader can trace the work using the bibliographic data supplied in the reference. It guarantees the authoritativeness of the work to some extent.

This is done by reporting the works of others in one's own words in the following ways:

Paraphrasing : this involves restating in writer's words what is written in the source.

Summarising: this involves condensing the essential or important ideas in the source in a few short statements.

Direct quotation: some parts of a writing may be quoted directly, but they must either be enclosed in quotation marks if they are short or indented if longer.

Dont load the paper with quotations, if more than a quarter of your essay or paper consists of quotations, the assessor might get the impression that the writer has no ideas of his own.

In all cases you need to acknowledge other people's work.

APA Referencing

Books

1. Weedon , C. (1999) Feminism, Theory and Politics of Difference, Oxford, Blackwell.
2. Ajala V. (2001). Public Relations: in Search of Professional Excellence 2nd edn. Ibadan, Maybest Publications.

Journals

1. Bassey B.A. (2006). User Satisfaction with Services in three Academic Libraries in Cross Rivers State: a Comparative Study. Gateway Library Journal 9 (2):21-29

Websites

1. Kinley, W. (2011). Public Relations Practices in American Libraries.<http://www.ter.org.uk/fs10/further.htm>. Accessed 15 December, 2011

E-journals

Miller,K. (2010). Environmental literacy and green volunteer opportunities for your community. Public libraries online 32 (3) . [http:// www.publiclibrariesonline.org](http://www.publiclibrariesonline.org). Accessed 13 October , 2010.

Using the Library

Rules and regulations: Read the Library Guide

Preservation of Library materials

The reason behind conservation and preservation practices of library materials is to protect the expensive resources and make them available to a good number users. Knowledge of conservation and preservation prolongs the life of materials and guides against deterioration.

Causes of damage to Library materials

- ☒ Mutilation: this refers to deliberate removal of parts of library materials by delinquent users. Instead of mutilation, it would be more responsible to photocopy or borrow
- ☒ Defacing: marking
- ☒ Natural disasters: flooding
- ☒ Dust, uncontrolled temperature, high humidity
- ☒ Excessive sunlight
- ☒ Pests : avoid eating in the library
- ☒ Excessive photocopying: weakens the spine
- ☒ Poor quality paper: high acidic content
- ☒ Folding

Conservation/ preservation practices

- ☒ Air conditioning in high temperature areas
- ☒ Ventilation
- ☒ Dusting
- ☒ Avoid eating in the library
- ☒ Fumigation to deal with pests
- ☒ Proper handling and storage
- ☒ Disaster management e.g. Insurance
- ☒ Reformation by photocopying, microfilming and digitisation
- ☒ User education

Library Tour/ Practical

As soon as lectures for this course are completed, class reps. should endeavor to meet with the lecturer and prepare a time table for library tour.